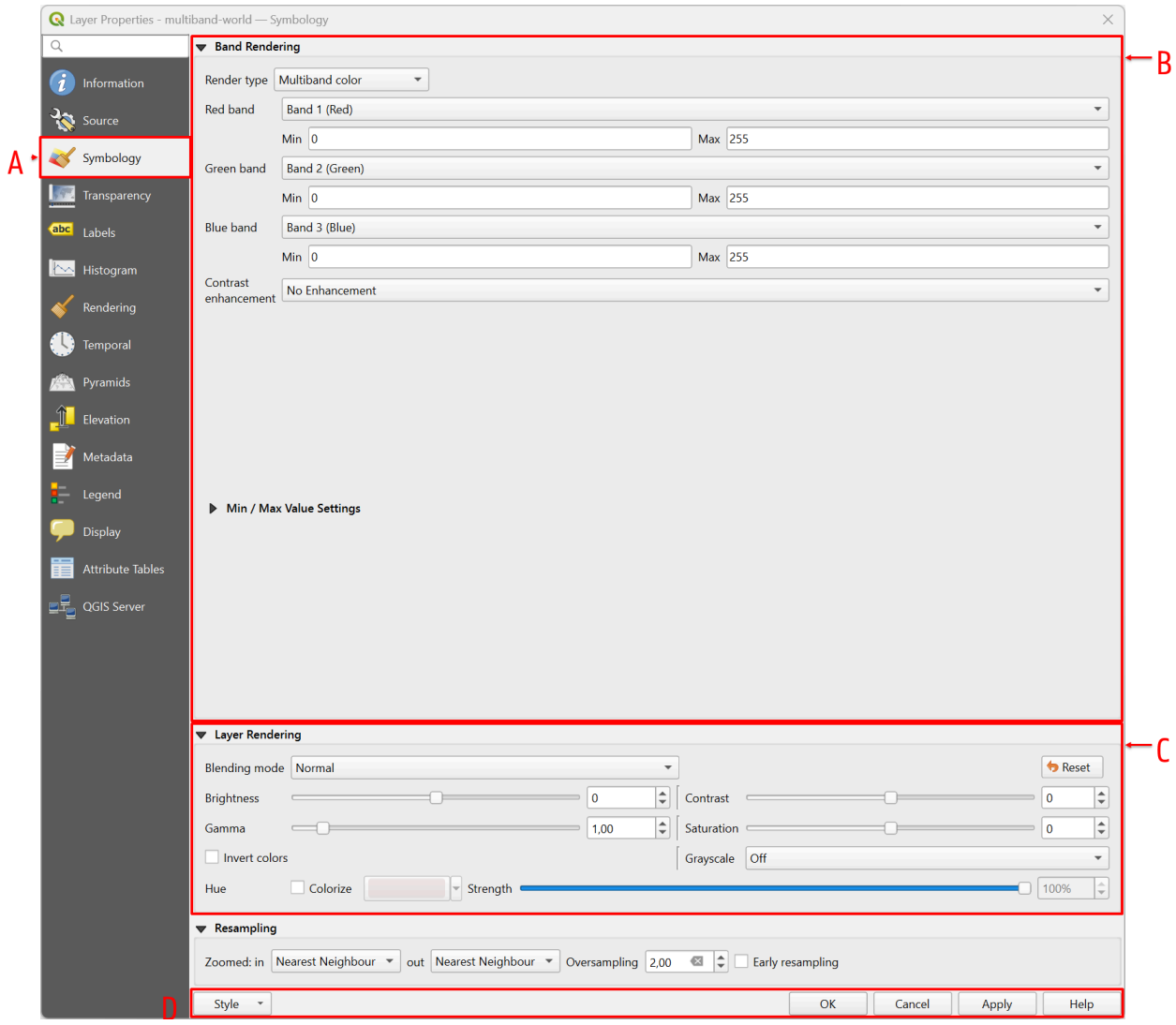
4.3.2 Visualising raster data

Changing the display style of a raster layer

General

You set how your raster is shown in the map canvas on the *Symbology* (A) tab.

- Under **Band Rendering** (B) you decide how the values in your raster cells are translated into colours. For this you choose a *render type*: the most common ones are explained below.
- Under **'Layer Rendering'** (C) you change the overall brightness, saturation or contrast of the colours in your raster layer.
- Click **'OK' or 'Apply'** (D) to apply your changes.



Render types

In 2.1 The raster model you learned that raster cells contain numeric values. What these values represent depends on the raster data. They may represent real-world quantities, such as elevation, or codes representing categories, such as soil types. In a scanned map or photograph, however, the values may represent colours rather than the real-world information shown in the image.

In the map canvas, you do not normally see the numeric values stored in the raster. Instead, each cell is displayed in a colour. To do this, the actual numeric values are translated into colours according to a set of display rules.

You can control how QGIS does this on the *Symbology* tab of *Layer Properties*. Under *Band Rendering*, you first choose a *render type* (display style). The appropriate choice depends mainly on the number of image bands in the raster and on the kind of data it contains. Below, we explain three common render types, followed by a short overview of the others.

Multiband color

In a colour photo or colour scan, the stored values represent colour information from the photo or scan itself. This information is stored in three image bands: one for red (R), one for green (G) and one for blue (B). Some images also have a fourth band, called the *alpha* band, which stores transparency information.

RGB values usually range from 0 to 255. For example, RGB (0, 255, 0) represents bright green, RGB (0, 0, 0) represents black, and RGB (255, 255, 255) represents white.

The *Multiband color* renderer combines the red, green and blue bands to display the image in colour. It also allows you to adjust the contrast of each band separately (B). The screenshot under *General* above shows these settings for a colour image of the world.

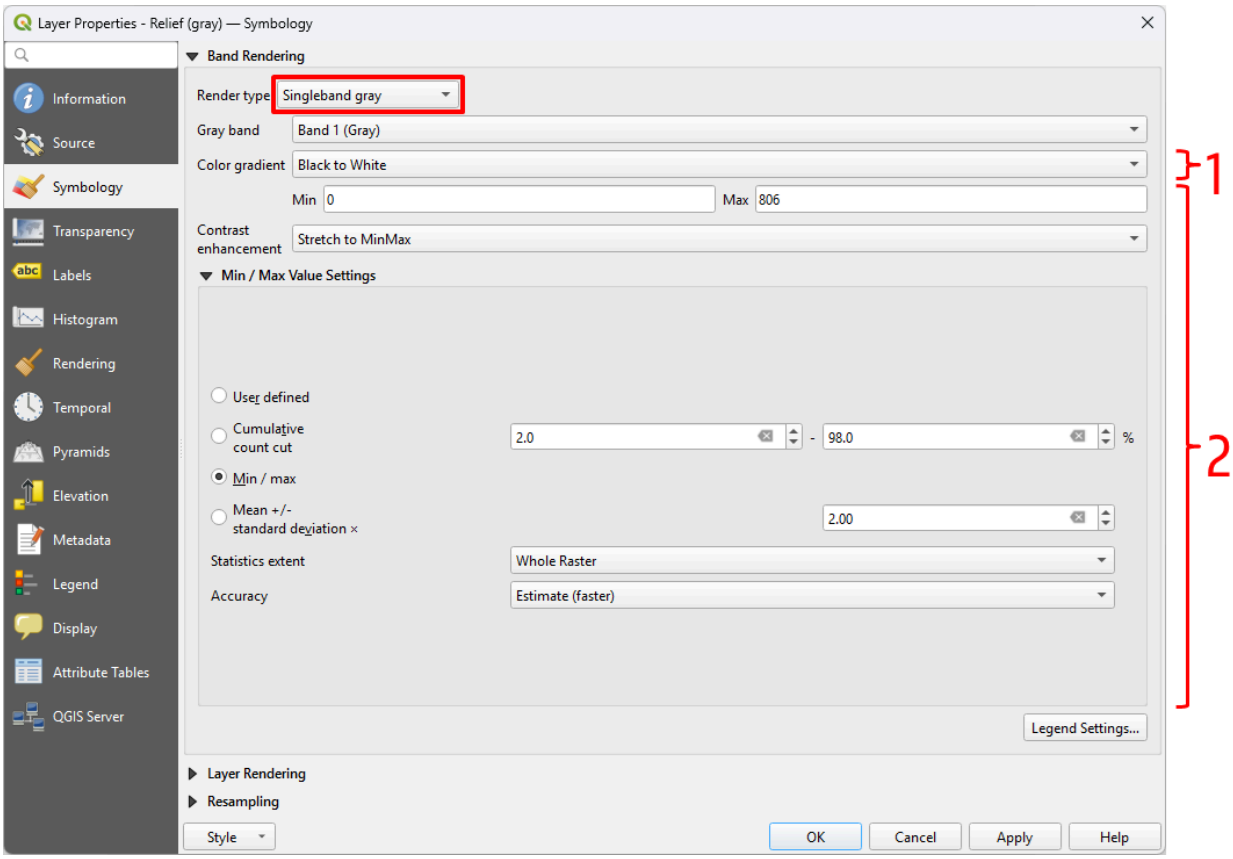
Singleband gray

If your raster layer consists of one image band, the render types *Singleband gray* and *Singleband pseudocolor* are often used. The examples below use a digital elevation model (DEM) of northern France and Belgium: each cell contains the height of the ground in metres.

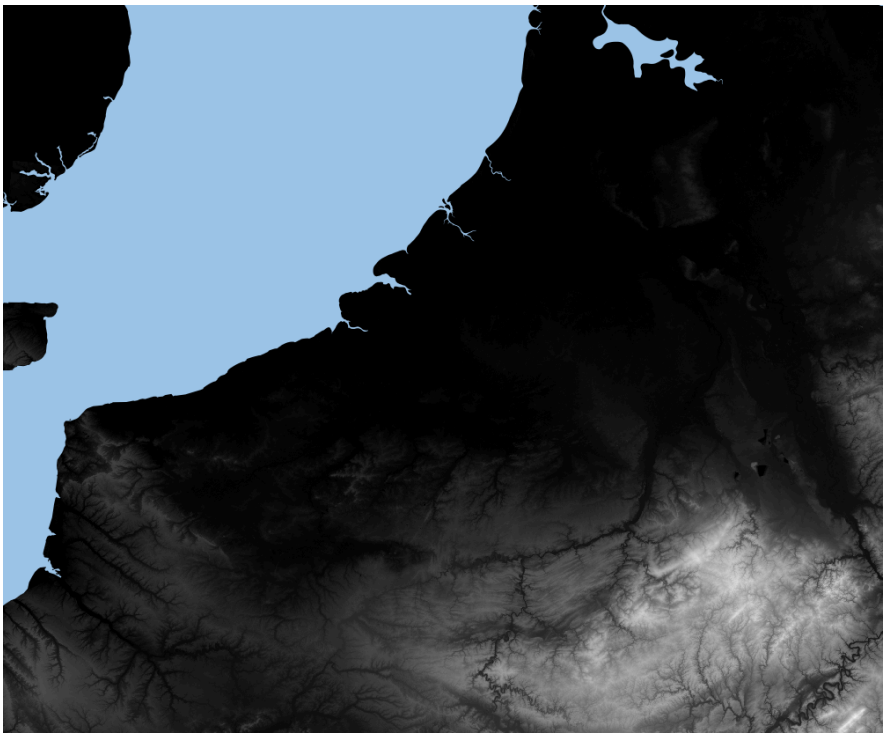
The render type *Singleband gray* displays cell values using shades of grey. Under *Color gradient*, you choose either *White to Black* (higher values are darker) or *Black to White* (higher values are lighter) (1).

Under *Contrast enhancement* and *Min / Max Value Settings* (2), you set the range of cell values used for the colour gradient. By default, QGIS uses *Stretch to MinMax* and *Min / max*. It determines the lowest and highest cell values in the raster and assigns them the two end colours of the gradient (black and white). It estimates these values from a sample of cells. Under *Accuracy*, you can choose *Actual (slower)* to use the actual minimum and maximum values.

Alternatively, you can choose *Cumulative count cut* (2–98%). This ignores the lowest 2% and highest 2% of values when determining the range, which often increases the contrast. In practice, you often **do not need to change** these defaults.



The map below shows the result with the settings in the screenshot (*Black to White*, minimum 0 m and maximum 806 m): the lowest areas are black, the highest white.



Elevation data: SRTM (NASA/USGS, public domain).

Singleband pseudocolor

The *Singleband pseudocolor* render type lets you display a single-band raster using colours other than shades of grey. You choose a *Color ramp* (1) and set the minimum and maximum values in the same way as for *Singleband gray* (2). However, this render type offers more flexibility in how colours are assigned to cell values.

You can divide the range of values into classes, either by setting the class boundaries manually or by using a classification mode and specifying the number of classes (4).

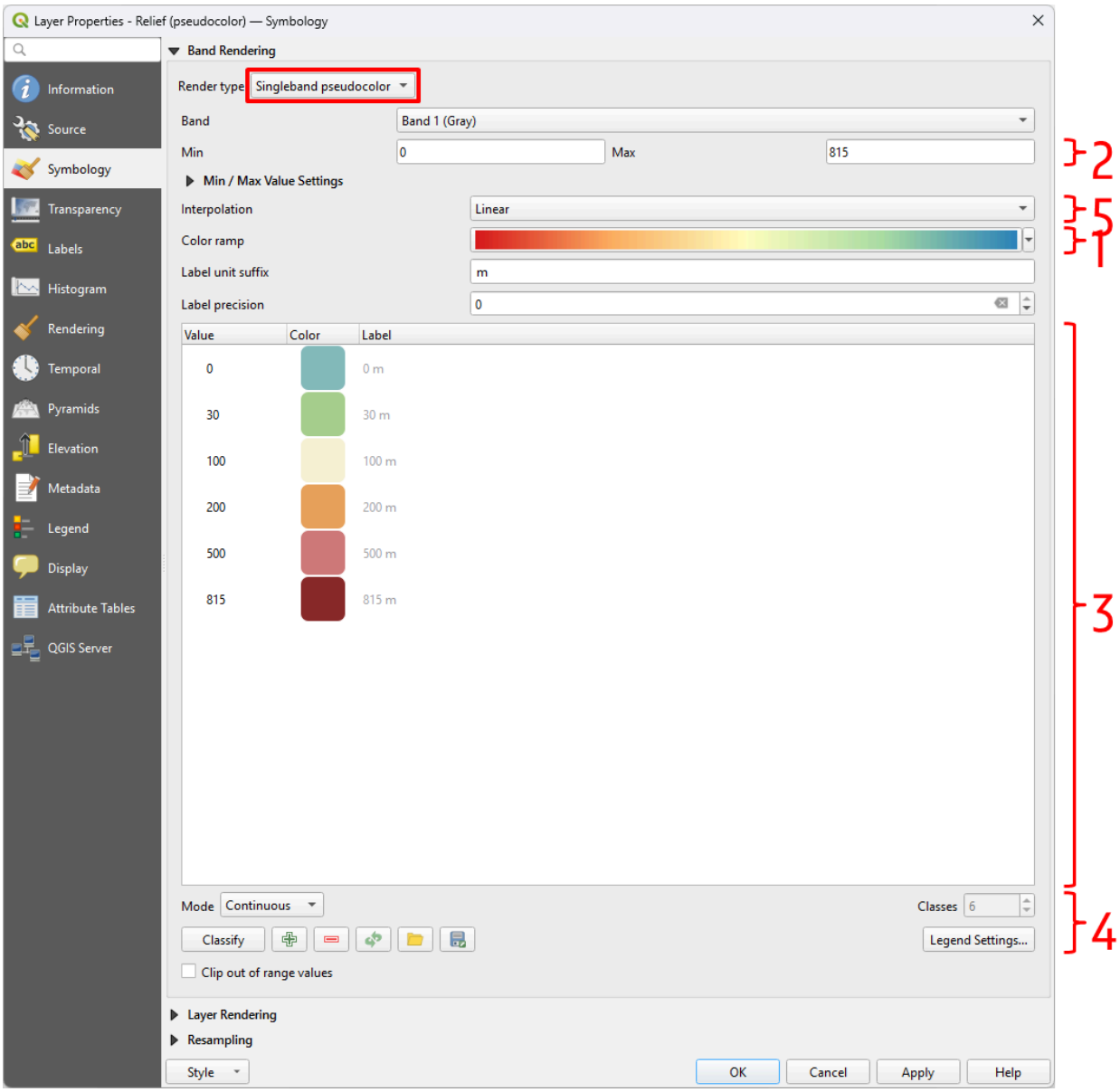
The available modes include:

- *Continuous* (the default): QGIS creates a reference value for each colour stop in the selected color ramp. The number of colour stops depends on the ramp. For example, the single-hue green, blue and red ramps have nine predefined colour stops, while ramps such as *Magma* have more than 50.
- *Equal Interval*: You specify the number of classes, and QGIS divides the range between the minimum and maximum values into intervals of equal size.
- *Quantile*: QGIS divides the cells into classes so that each class contains approximately the same number of cells.

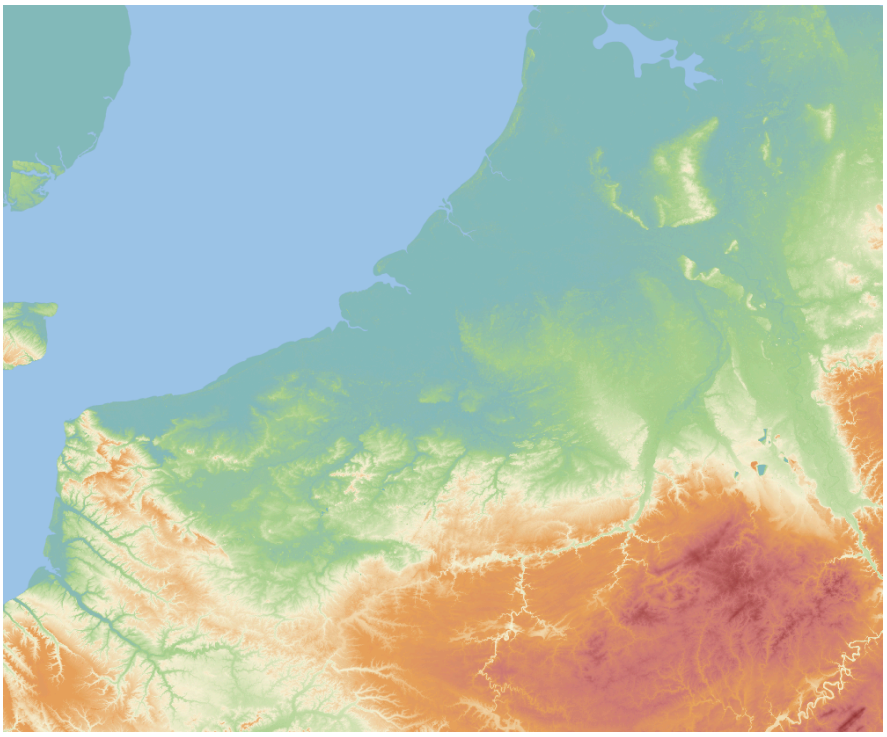
The table (3) lists the reference values (*Value*) and their colours (*Color*). Click *Classify* to let QGIS calculate the values with the chosen mode, colour ramp and number of classes, or set the values and their colours yourself: double-click a value or colour to change it, and add or remove rows with the plus and minus buttons below the table.

In addition to defining the classes, this renderer lets you choose how cell values within each class are coloured. You do this using the interpolation method (5):

- *Discrete*: Each class is displayed using a single colour, determined by the upper boundary of the class. This works in the same way as *Graduated* symbology for vector layers.
- *Linear*: QGIS creates a colour ramp between the colours assigned to the lower and upper boundaries of each class. Colours change gradually along this ramp, so cells with values between the boundaries receive intermediate colours.
- *Exact*: Only cells whose values match those listed in the table are coloured. Cells with other values receive no colour and are displayed as transparent.



In the example above, the reference values (0, 30, 100, 200, 500 and 815 m) are set manually to emphasise differences in elevation. The lower areas are shown in several shades of green, while the higher terrain stands out in shades ranging from orange to red.



Elevation data: SRTM (NASA/USGS, public domain).

Other render types

QGIS 4 has a few more render types, among others:

- *Paletted/Unique values* gives each unique cell value its own colour. It suits discrete data, such as the codes of land use or soil types.
- *Single color* shows all cells that contain data in one colour, for example to show only where a raster has data.
- *Hillshade* adds shading, so that the values look like three-dimensional relief. It is mostly used for elevation models.
- *Contours* displays contour lines. It is also mostly used for elevation models.

The most suitable render type and its settings depend on the technical nature of your raster, but also on the type of data it stores.

> Tip
>
> Use a colour gradient for a raster layer with **continuous data** (temperature, elevation…). If your raster contains **discrete data**, give each data class a clearly different colour, for example with the render type *Paletted/Unique values*, or with *Singleband pseudocolor* and *Discrete* or *Exact* interpolation. (Continuous and discrete data are explained in Raster model.)

NoData

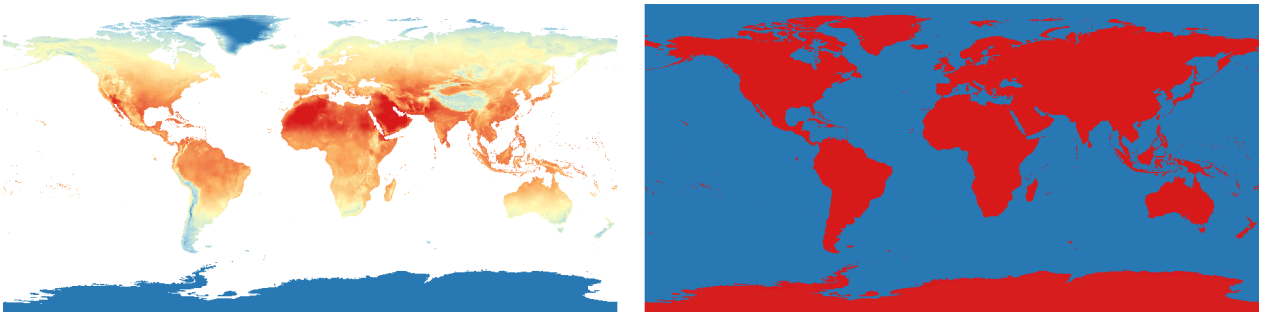
Not every cell of a raster contains data. In a temperature raster, for example, there are no measurements for sea areas. Such cells get a special value that the file marks as **NoData**.

Often, the value representing NoData is a number that cannot occur in the data, such as –9999. Usually, the information identifying NoData cells is stored in the raster file's metadata, and QGIS recognises it

automatically. When QGIS correctly recognises cells as NoData, it does not draw them and excludes them when calculating the minimum and maximum of the raster.

Sometimes, however, a raster comes without this information. QGIS then treats −9999 erroneously as a normal value: the cells are drawn and affect the colour range drastically. With *Min / max*, the colour gradient then runs from −9999 to the actual maximum, so all valid values end up in almost the same colour.

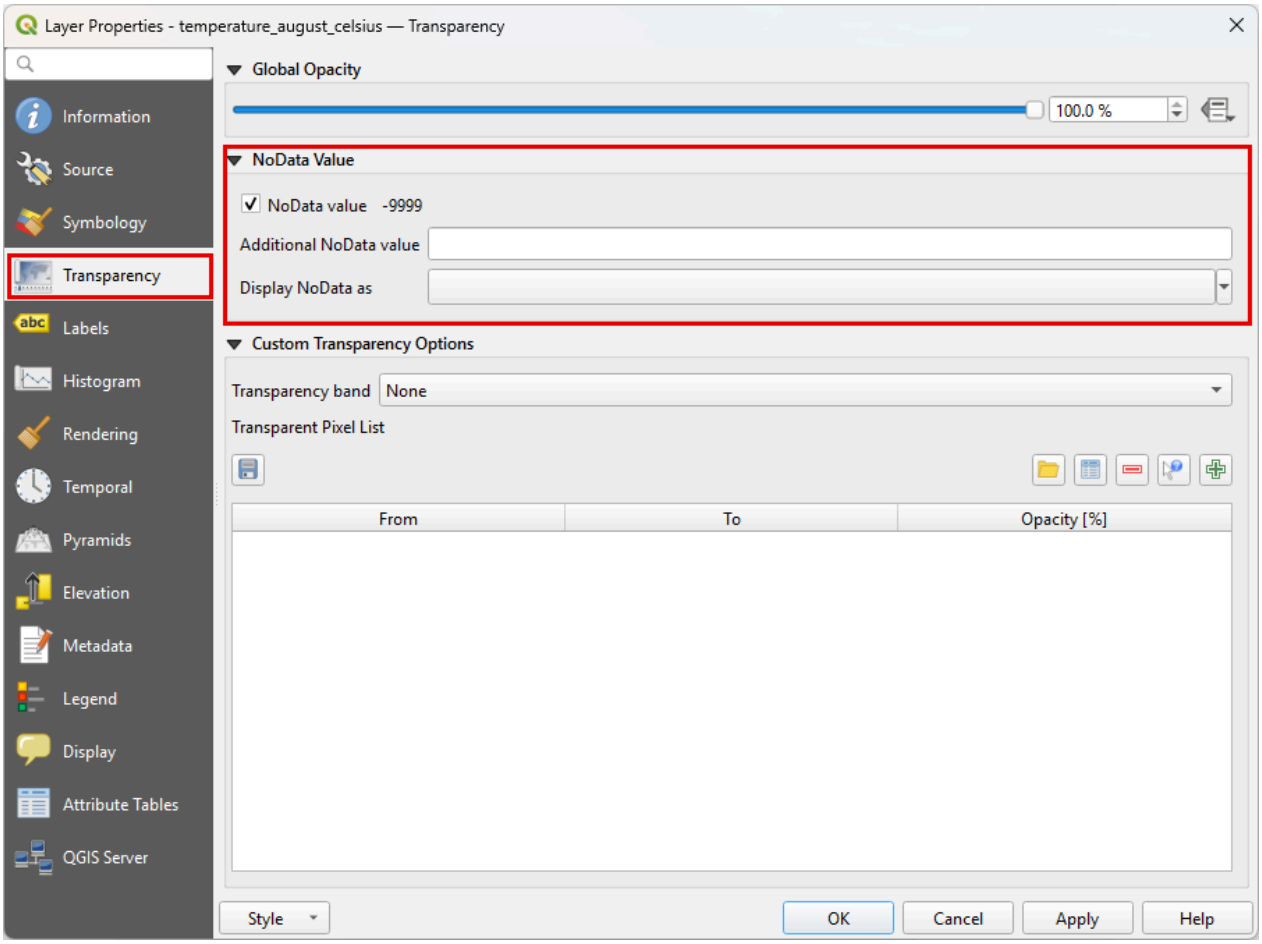
The images below show the same temperature raster, on the left with its NoData value and on the right without it.



Left: the sea is NoData and is not drawn. Right: the sea has the value −9999, and all land gets the same colour. Temperature data: WorldClim 2.1 (CC BY 4.0).

You check and set the NoData value in the *Layer Properties*, on the *Transparency* tab, under *NoData Value*:

- *NoData value* shows the value that is stored in the file. If you untick it, QGIS no longer treats these cells as NoData.
- Under *Additional NoData value*, you enter extra values that QGIS should treat as NoData, for example −9999 for a raster that comes without NoData information.



Tip

Does your raster show strange colours, or a minimum value such as –9999? Check the NoData value first.